



Training & Skilling Program in Space Technology

Empowering Future Space Scientists

EARTH NOW
SPACE AGRONOMY AS SERVICE

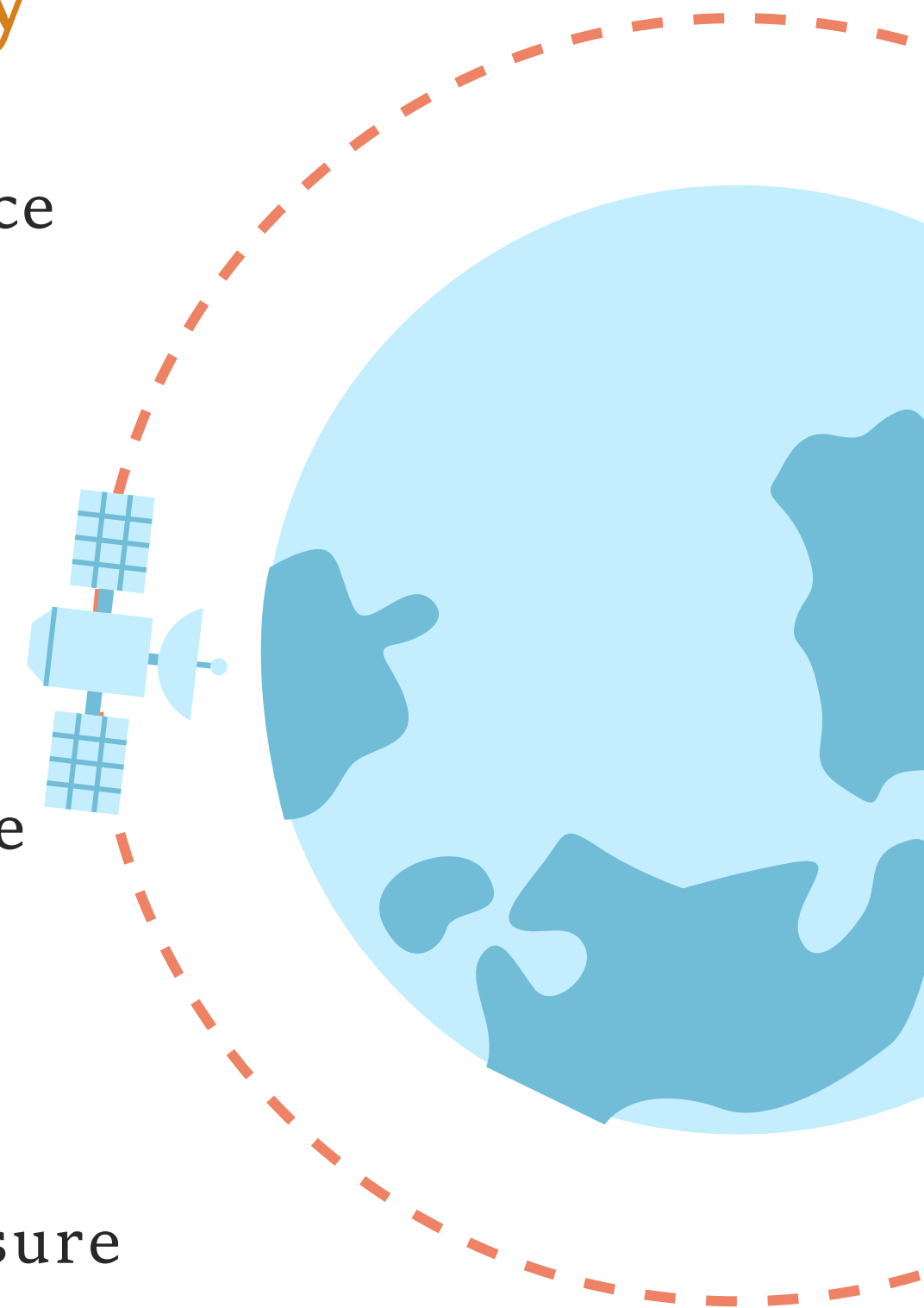

SMARTCIRCUITS[®]
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 TAKEME2SPACE



Igniting Innovation Through Hands-On Skilling in Space Technology

- A national skilling initiative led by Earth Now Pvt. Ltd, TM2Space Technologies Pvt Ltd, and Smart circuits Innovation Pvt. Ltd.
- Offers tiered learning:
 - 1-Week Nano Satellite Bootcamp (Manha Kit)
 - 8-12 Week Advanced Satellite Program
 - Capstone Project with ISRO POEM Missions
- Focused on hands-on learning, real payload design, and satellite prototyping
- Aligned with Atmanirbhar Bharat, empowering students with industry-relevant skills
- Open to schools and colleges, with or without prior STEM exposure



Powerful Collaboration for National Space Talent Development

- A unique partnership between three pioneers in space education and innovation:



Earth Now

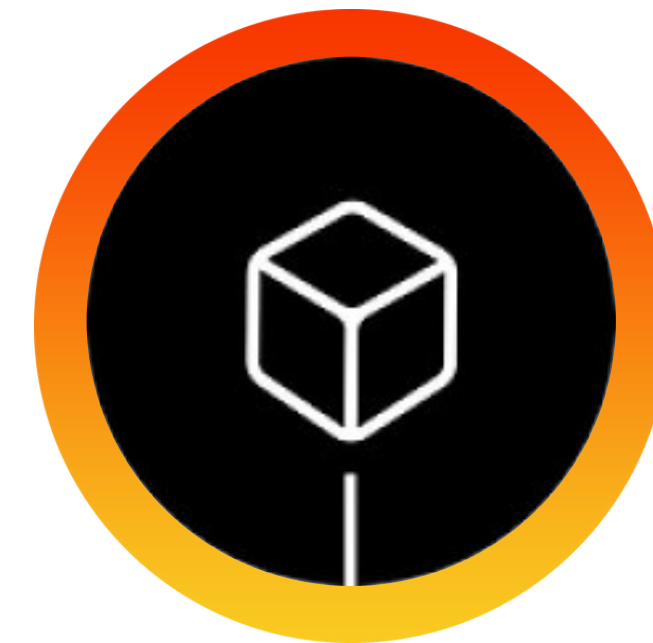
Specialists in satellite data analytics and Earth observation solutions.



Smartcircuits

Innovation

ISRO-certified Space Tutor, focused on educational outreach and hands-on training programs.



TakeMe2Space

Technical experts in satellite R&D, with experience in conducting real space missions.

- Together, we deliver cutting-edge space skilling programs from design to data analysis.
- Aimed at creating a space-literate, innovation-driven youth workforce across India

Creating Future-Ready Space Innovators for India

Why This Skilling Program is Important?

- India's space sector is booming – with growing demand for skilled talent in satellite technology, data analytics, and mission operations
- Students need real exposure – not just theory, but hands-on experience in satellite design, testing, and even launching
- Bridges the gap between curriculum and career – prepares students for emerging roles in aerospace, AI, electronics, and Earth observation
- Aligned with national priorities – supports Atmanirbhar Bharat, ISRO's student programs (POEM, SMiLE), and NEP 2020 objectives
- Encourages innovation & curiosity – empowers students to design solutions for climate, agriculture, disaster response, and more
- Equips the next generation – to lead India's journey in becoming a global space superpower

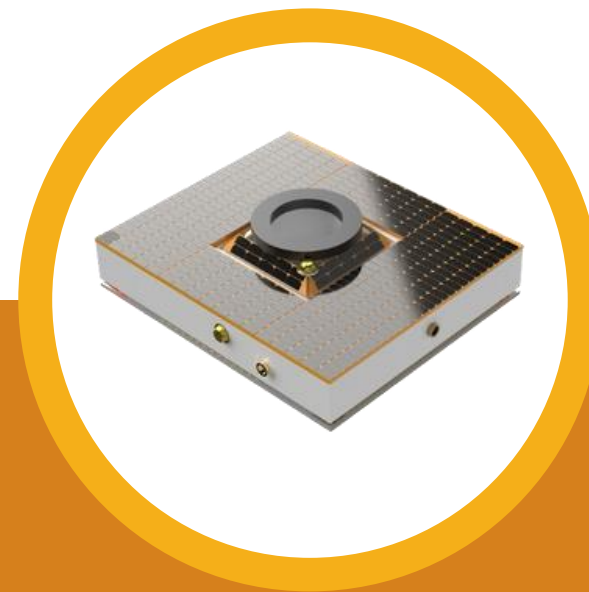


Skilling Program Modules



Short-Term
1 Week

Nano satellite development
(Manha Kit)



Mid-Term
12 Weeks

Advanced satellite systems,
payloads, and telemetry



Capstone
Project

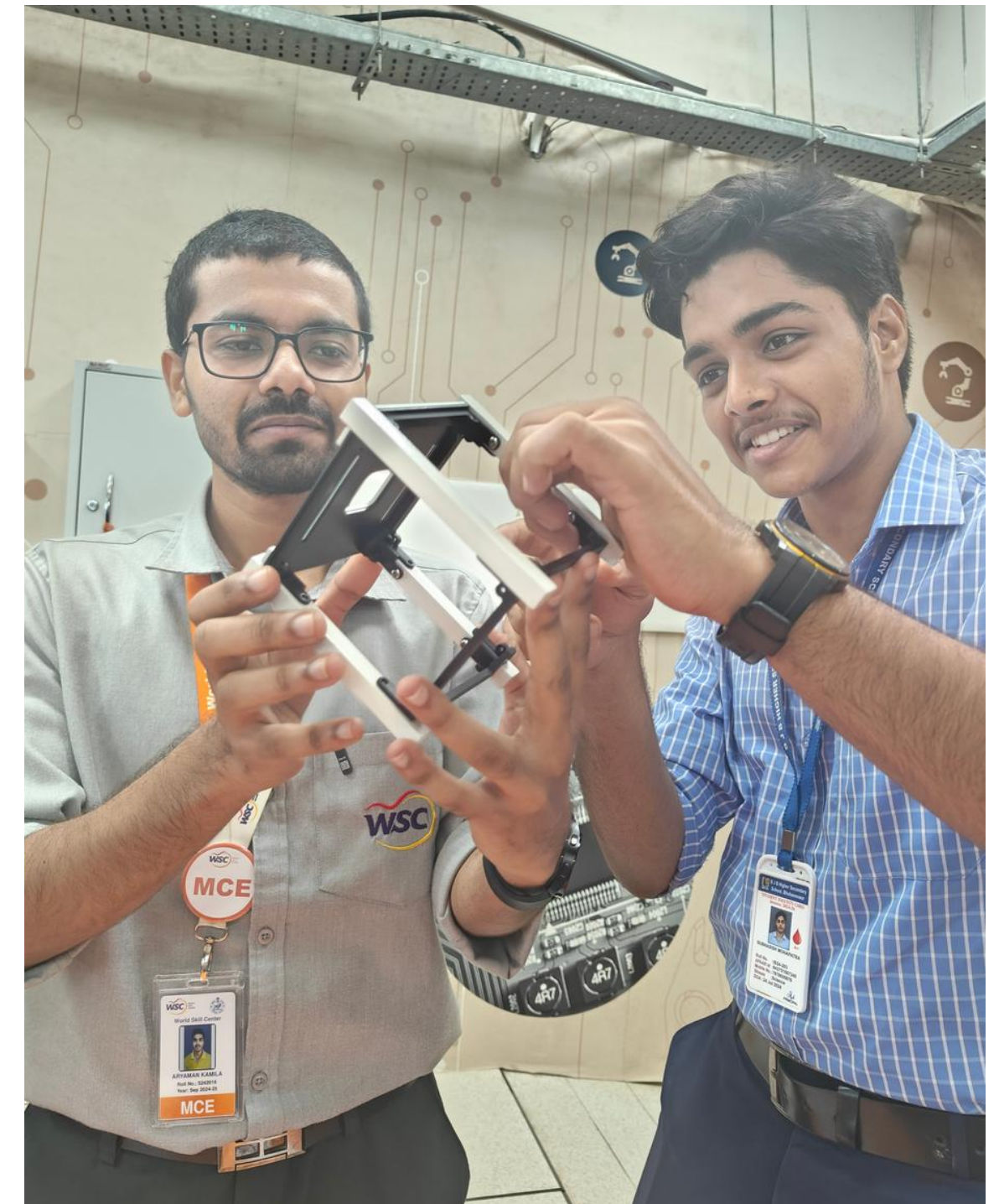
Real payload development
for ISRO's POEM mission

- Programs designed for students from schools, colleges, and technical institutions.
- Curriculum aligned with ISRO, IN-SPACE, and Atmanirbhar Bharat space initiatives.
- Focused on hands-on learning, mission simulation, and innovation.

Nano Satellite Development Bootcamp (1 Week)

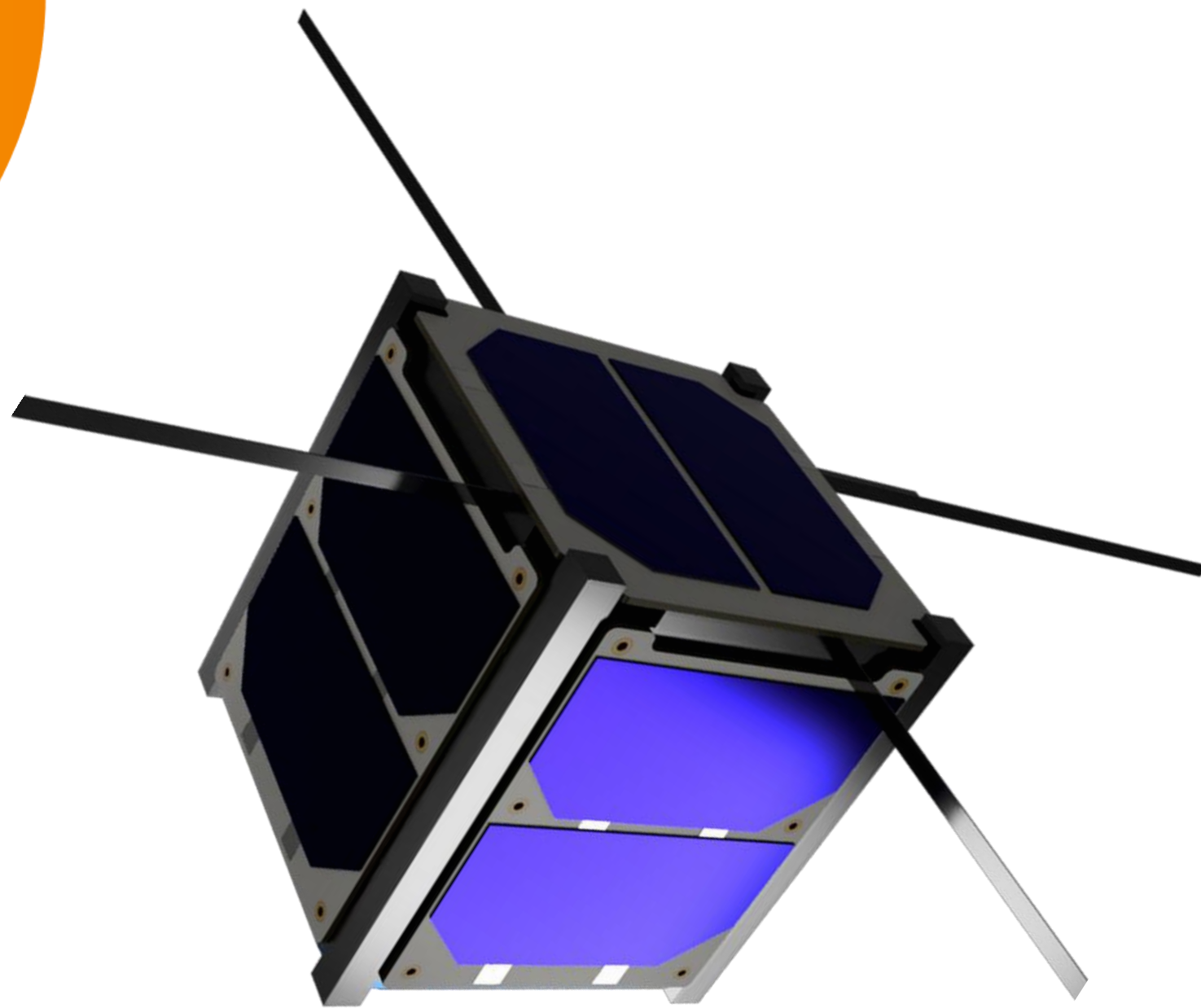
Making a Nano Satellite using MANHA Sat Kit

- Introduction to Satellite Concepts – Types, orbits, rocket science, and ground stations
- Hands-on Kit Training – Learn electronics, sensors, microcontrollers, and circuitry
- Live Signal Reception – Track and decode NOAA satellite data using ground station
- Mechanical Assembly – Build the satellite structure and integrate components
- Programming & Testing – Code and upload software onto the satellite kit
- Parachute Design – Learn descent mechanics and conduct drop tests
- Drone Launch & Data Analysis – Launch the satellite and analyze real-time data
- Teamwork & Certification – Team-based project execution and certificate distribution



From Classroom to CubeSat Prototype

12 Week Advanced Training Program

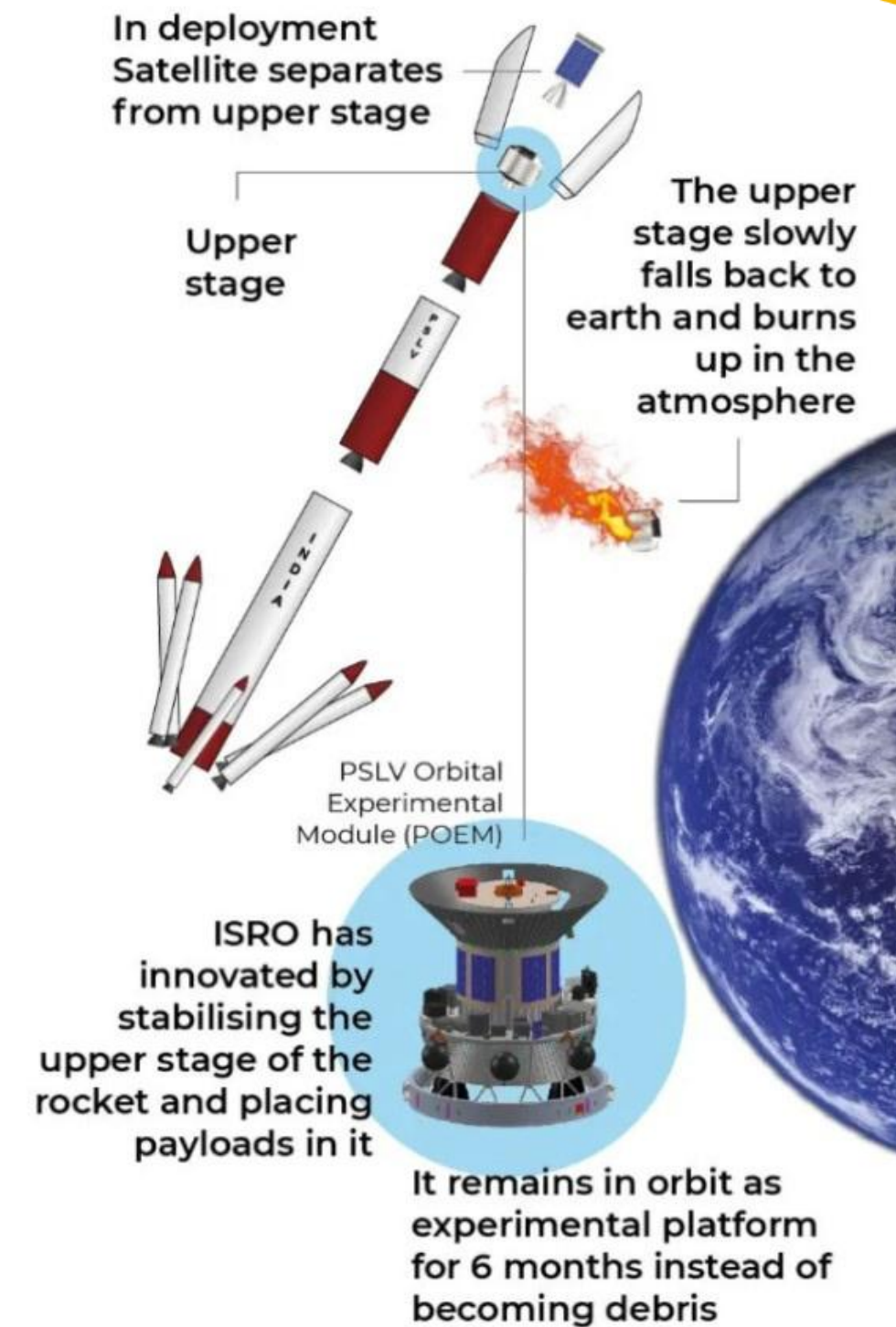


- In-depth Satellite Systems Training – Learn about OBC, EPS, ADCS, payload, and communication modules
- Hands-on Prototyping – Build a functional Flatsat setup with real satellite subsystems
- Mission Design & Simulation – Use tools like STK, MATLAB, and Gpredict for orbit simulation
- Embedded Systems Programming – Code using C/C++, Python, and microcontroller environments
- Subsystem Testing – Validate components using logic analyzers, multimeters, and oscilloscopes
- Mini Ground Station Setup – Learn basics of satellite tracking and data reception
- Team-Based Projects – Design custom satellite missions with mentoring support
- Capstone Review & Presentation – Demonstrate working model and present mission plan

ISRO POEM/SMiLE Missions

Real Payload Development for National Space Missions

- Work on Real Payloads – Design and develop functional experiments for ISRO POEM or SMiLE platforms
- Mission Lifecycle Experience – Go through concept, design, documentation, and testing phases
- ISRO/IN-SPACE Engagement – Participate in payload selection, review, and validation stages
- Rigorous Testing Process – Undergo ISRO-specified tests: Vibration, TVAC, EMI/EMC, Shock
- Integration and Launch Support – Visit facilities like VSSC and SHAR for integration & launch
- Mentorship by Experts – Get guidance from aerospace engineers and mission professionals
- National Recognition – Be part of India's student satellite mission ecosystem
- Career-Boosting Exposure – First-hand experience in contributing to India's space program



Building Confidence, Capability, and Curiosity

Training Outcomes



- Hands-on Learning – Students build, test, and launch satellite models
- Real-World Skills – Exposure to embedded systems, coding, electronics, and satellite communication
- Mission Experience – Opportunity to work on ISRO-linked payloads and participate in national-level space missions
- Career Awareness – Understanding roles in aerospace, electronics, data science, and space research
- Boost in Innovation & Teamwork – Collaborative learning fosters creative thinking and leadership
- Portfolio Building – Students create tangible project outcomes to showcase in future academic or career opportunities

Empowering India's Next-Gen Space Innovators

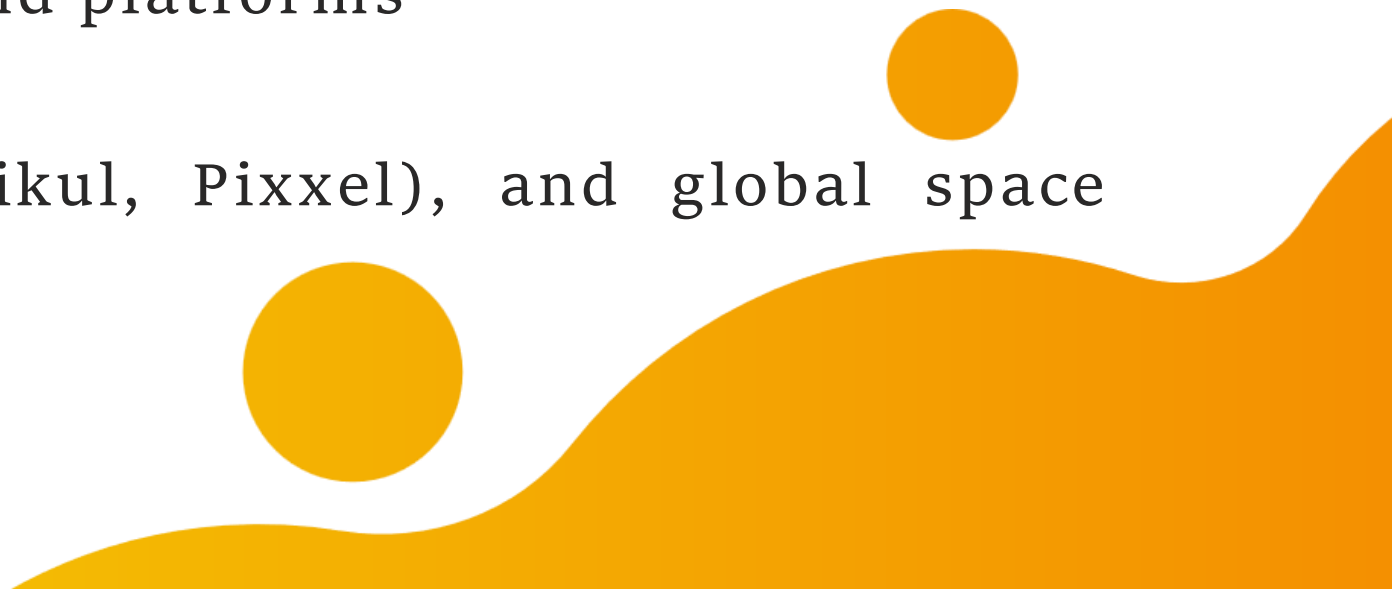
Target Audience & Reach



- Designed for school students (Grade 6–12), polytechnic, and engineering college students
- Suitable for urban, semi-urban, and rural learners—no prior technical background required
- Successfully implemented in multiple states with diverse institutions and regional coverage
- Programs delivered in multiple languages and formats (offline/online/hybrid)
- Special focus on Tier-2 and Tier-3 cities, expanding access to quality space education
- Reached over 1,00,000+ students through past workshops, missions, and training programs

From Skilling to Space Careers – Your Journey Begins Here

Career & Future Opportunities

- Junior Space Technologist / Remote Sensing Analyst
 - Entry-level roles in satellite operations, data acquisition, and Earth observation projects
 - Space Systems Engineer / Mission Planner
 - Design, simulate, and plan satellite missions for government and private space agencies
 - Research Engineer – Aerospace/Space Applications
 - Innovate in propulsion, satellite subsystems, onboard software, and sensor technologies
 - Other Emerging Career Paths:
 - Embedded Systems Developer for space and defense
 - Geospatial Data Analyst using AI/ML on satellite imagery
 - CubeSat Developer or Payload Integration Specialist
 - Space Educator or STEM Trainer in schools and innovation labs
 - Entrepreneur in space-tech or EdTech using satellite kits and platforms
 - Opportunities with:
 - ISRO, IN-SPACe, DRDO, private startups (Skyroot, Agnikul, Pixxel), and global space agencies (NASA, ESA)
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Thank You

*"Look at the sky. We are not alone.
The whole universe is friendly to
us and conspires only to give the
best to those who dream and
work."*

— Dr. A.P.J. Abdul Kalam